26900103 **Reg.No : .....**

Name : .....

MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU-UGP (Honours)/ MGU-BBA (Honours)/ MGU-BCA (Honours)

REGULAR EXAMINATION October 2026

Fifth SEMESTER

Discipline Specific Core Course (DSC) - MG5DSCCHE301 - PHYSICAL CHEMISTRY 2

(2024 ADMISSION ONWARDS)

Duration: 1 hour 30 minutes **Maximum Marks: 50**

Remember(K), Understand(U), Apply(A), Analyse(An), Evaluate(E), Create(C), Skill(S), Interest(I) and Appreciation(Ap)

Students should attempt at least one question from each course outcome to enhance their overall outcome attainability.

Part A

Multiple Choice Questions

Answer all questions

Each question carries **1** marks

1. X-ray diffraction by crystals is mainly used to determine: [U] / [CO1]

   a). Melting point b). Molecular mass

   c). Crystal structure d). Boiling point

2. Silicon doped with arsenic produces a: [K] / [CO1]

   a). p-type semiconductor b). n-type semiconductor

   c). Metallic conductor d). Superconductor

3. In the smectic phase, the molecules are: [K] / [CO1]

   a). Randomly oriented b). Arranged in layers with their long axes approximately parallel

c). Arranged only in cylindrical structures

d). Completely rigid

4. In the photosensitised isomerisation of 2-butene, the sensitizer [U] / [CO2]

a). Absorbs light and transfers excitation energy to 2-butene

b). Is permanently consumed in the reaction

c). Acts only as a solvent

d). Prevents excitation of 2-butene

5. In a Jablonski diagram, the lowest electronic singlet state is represented as: [U] / [CO2]

a). S_0 b). S_1 c). T_1 d). T_2

6. For an isochoric process, the internal energy change is equal to [U] / [CO3]

a). Zero b). V

c). q_V d). $P\Delta V$

7. When work is done on a system under adiabatic condition, its internal energy [An] / [CO3]

a). does not change b). increases

c). decreases d). may increase or decrease

8. The area enclosed by a Carnot cycle on a P–V diagram represents [U] / [CO4]

a). Heat supplied b). Heat rejected

c). Net work done d). Change in internal energy

9. When a real gas shows a cooling effect during expansion through a porous plug, the Joule Thomson coefficient is [U] / [CO3]

a). Positive b). negative

c). zero d). Infinity

10. A substance having disorder or degeneracy even at 0 K may possess [U] / [CO4]

a). Zero entropy b). Infinite entropy

c). Residual entropy d). Negative entropy

[1x10 = 10]

Part B

Short Answer Questions

Answer any **4** questions

Each question carries **5** marks

11. Briefly explain the structure of zinc blende. [U] / [CO1]

12. Explain the Schottky defect. [K] / [CO1]

13. Calculate the energy per mole of light having wavelengths of (a) 85 nm and (b) 300 nm [U] / [CO2]

14. What are primary and secondary processes in photochemical reactions? [U] / [CO2]

15. State and explain the different statements of the second law of thermodynamics. [U] / [CO3]

16. Derive the expression for entropy as a function of volume and temperature. [A] / [CO4]

[5x4 = 20]

Part C

Short Essay

Answer any **2** questions

Each question carries **10** marks

17. (a) Explain in detail the law of rational indices. (b) Write a short note on Weiss indices and Miller indices. [U] / [CO1]

18. (a) Derive the expression for maximum work obtained during the reversible expansion of an ideal gas from V_1 to V_2 at constant temperature. (b) One mole of an ideal gas expands reversibly and isothermally at 300 K from 5 L to 20 L. Calculate the maximum work done. [K] / [CO3]

19. Derive Gibbs-Helmholtz equation. [U] / [CO4]

20. Derive Gibbs-Duhem equation. [U] / [CO4]

[10x2 = 20]